ADDITIVITY OF THE CHERN CHARACTER ON A SHORT EXACT SEQUENCE OF FINITE LOCALLY FREE SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The splitting principle, which makes the Chern roots of a finite locally free module available. Same chapter chow.tex, Section \ref{section-splitting-principle}, invoked at proof line 8976. It reduces the computation to a situation where the Chern roots $x_{1,i}$ of $\mathcal{E}_1$ and $x_{2,i}$ of $\mathcal{E}_2$ can be named.
- The polynomials P_p used to define the Chern character. Same chapter chow.tex, Example \ref{example-power-sum}, cited at statement line 8971 and proof line 8982. Their choice on a disjoint union of roots is exactly what turns the concatenation of the two root lists into the sum $P_p(\mathcal{E}_1) + P_p(\mathcal{E}_2)$.

Source reference. Original reference 0F9B: sources/stacks-chow/pinned-original-source.tex, source lines 8673–8673. Navigation pointer for \ref{example-power-sum} used inside the counted span (line 8971); the argument that consumes it is retained in the span.

Source reference. Original section 02UK: sources/stacks-chow/pinned-original-source.tex, source lines 8422–8422. Navigation pointer for \ref{section-splitting-principle} used inside the counted span (line 8976); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup}

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used inside the counted span (line 8959); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $0 \to \mathcal{E}_1 \to \mathcal{E} \to \mathcal{E}_2 \to 0$ be a short exact sequence of finite locally free $\mathcal{O}_X$-modules. Then we have the equality*

$$ch(\mathcal{E}) = ch(\mathcal{E}_1) + ch(\mathcal{E}_2)$$

More precisely, we have $P_p(\mathcal{E}) = P_p(\mathcal{E}_1) + P_p(\mathcal{E}_2)$ in $A^p(X)$ where P_p is as in Example 0F9B.

Proof. It suffices to prove the more precise statement. By Section 02UK this follows because if $x_{1,i}$, $i = 1, \ldots, r_1$ and $x_{2,i}$, $i = 1, \ldots, r_2$ are the Chern roots of $\mathcal{E}_1$ and $\mathcal{E}_2$, then $x_{1,1}, \ldots, x_{1,r_1}, x_{2,1}, \ldots, x_{2,r_2}$ are the Chern roots of $\mathcal{E}$. Hence we get the result from our choice of P_p in Example 0F9B. □